

EX:01	Design and Evaluate Different Prompt Types to Control Agent Reasoning, Autonomy, and Output Quality
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Aim:

To design and evaluate different prompt engineering techniques for controlling an AI agent's reasoning process, autonomy, and output quality, and compare their effectiveness on the same task.

Procedure:

1. Select a common task (e.g., text summarization, question answering, or explanation).
2. Design different prompt types:
 - Zero-Shot Prompt
 - Few-Shot Prompt
 - Role-Based Prompt
 - Chain-of-Thought Prompt
3. Provide the same input to the AI model using each prompt.
4. Collect the generated responses.
5. Evaluate each response based on:
 - Accuracy
 - Reasoning Quality
 - Completeness
 - Output Format
 - Response Length
6. Compare the outputs and record observations.
7. Conclude which prompt type performs best for the selected task.

Code:

```
from google import genai
```

```
client = genai.Client(api_key="YOUR_GEMINI_API_KEY")
```

```
task = "Explain why renewable energy is important."
```

```
prompts = {  
    "Zero-Shot":  
        task,
```

```
    "Few-Shot":  
        ""
```

Example:

Question: What is Artificial Intelligence?

Answer: Artificial Intelligence enables computers to perform tasks that normally require human intelligence.

Now answer:
Explain why renewable energy is important.
""",

"Role-Based":
"""

You are an environmental scientist.
Explain why renewable energy is important.
""",

"Chain-of-Thought":
"""

Think step by step before answering.

Question:
Explain why renewable energy is important.
"""
}

Generate responses for each prompt
for prompt_type, prompt in prompts.items():

```
    response = client.models.generate_content(  
        model="gemini-2.5-flash",  
        contents=prompt  
    )
```

```
    print("=" * 60)  
    print(prompt_type)  
    print("=" * 60)  
    print(response.text)
```

```
    print()
```

Output:

=====

Zero-Shot

=====

Renewable energy is crucial for a multitude of interconnected reasons, impacting our environment, health, economy, and global stability. Here's a breakdown of why it's so important:

1. ****Combating Climate Change:****

* ****Reduced Greenhouse Gas Emissions:**** The burning of fossil fuels (coal, oil, natural gas) releases vast amounts of greenhouse gases (like carbon dioxide and methane) into the atmosphere, trapping heat and causing global warming. Renewable energy sources (solar, wind, hydro, geothermal) produce little to no greenhouse gas emissions during operation, making them key to decarbonizing our energy systems and mitigating climate change impacts.

2. ****Improving Air Quality and Public Health:****

* **Less Air Pollution:** Fossil fuel combustion releases harmful pollutants like particulate matter, sulfur dioxide, nitrogen oxides, and mercury. These pollutants cause respiratory diseases (asthma, bronchitis), heart conditions, cancers, and premature deaths. Renewable energy dramatically reduces these emissions, leading to cleaner air and significant public health benefits, lowering healthcare costs.

3. **Enhancing Energy Security and Independence:**

* **Reduced Reliance on Finite Resources:** Fossil fuels are finite resources concentrated in specific regions, making supply vulnerable to geopolitical instability, price volatility, and supply disruptions. Renewable energy sources are abundant and widely distributed, allowing countries to generate power domestically and reduce dependence on foreign energy imports.

* **Stable Energy Prices:** Once built, renewable energy facilities primarily harness free "fuel" (sunlight, wind). This leads to more predictable and stable energy costs over the long term, less susceptible to global market fluctuations.

4. **Economic Growth and Job Creation:**

* **New Industries and Jobs:** The renewable energy sector is a rapidly growing industry creating millions of jobs globally in manufacturing, installation, operation, maintenance, research, and development. These jobs are often local and cannot be outsourced.

* **Investment Opportunities:** Significant investments are being made in renewable energy infrastructure, fostering economic growth and technological innovation.

* **Cost Reductions:** The cost of renewable technologies (especially solar and wind) has plummeted over the past decade, making them increasingly competitive with, and often cheaper than, fossil fuel alternatives.

5. **Sustainable Development for Future Generations:**

* **Infinite Resources:** Unlike fossil fuels, which are depleted with use, renewable resources are naturally replenished on a human timescale. This ensures a sustainable energy supply for future generations without exhausting finite planetary resources.

* **Environmental Stewardship:** By minimizing pollution and resource depletion, renewable energy supports a more sustainable relationship between human society and the natural environment.

6. **Decentralization and Resilience:**

* **Distributed Generation:** Many renewable energy systems can be decentralized, meaning power can be generated closer to where it's consumed (e.g., rooftop solar). This can increase grid resilience, reduce transmission losses, and provide power to remote areas not connected to central grids.

* **Reduced Vulnerability:** A diversified energy mix with a strong renewable component makes a nation's energy system more robust and less vulnerable to single points of failure or attacks.

In essence, renewable energy is important because it offers a pathway to a cleaner, healthier, more secure, and economically prosperous future for everyone, without compromising the ability of future generations to meet their own needs.

Few-Shot

Answer: Renewable energy is important because it offers a sustainable power source that helps combat climate change and reduces our reliance on finite fossil fuels.

Role-Based

As an environmental scientist, I can confidently state that the transition to renewable energy sources is not merely beneficial, but **critically important** for the long-term health of our planet, our economies, and human societies. Here's a breakdown of the key reasons:

1. **Mitigating Climate Change:**

* **The Problem:** The combustion of fossil fuels (coal, oil, natural gas) for energy production is the single largest contributor to greenhouse gas (GHG) emissions, primarily carbon dioxide (CO₂), but also methane and nitrous oxide. These GHGs trap heat in the Earth's atmosphere, leading to global warming and the cascading effects of climate change: sea-level rise, extreme weather events (heatwaves, droughts, floods, intense storms), ocean acidification, and disruption to ecosystems and biodiversity.

* **The Solution:** Renewable energy sources (solar, wind, hydropower, geothermal, biomass) produce little to no GHG emissions during operation. Shifting to these sources is the most direct and effective way to decarbonize our energy systems and avoid the most catastrophic scenarios of climate change.

2. **Improving Air Quality and Public Health:**

* **The Problem:** Beyond GHGs, burning fossil fuels releases a cocktail of harmful air pollutants, including particulate matter (PM_{2.5}), sulfur dioxide (SO₂), nitrogen oxides (NO_x), carbon monoxide (CO), and volatile organic compounds (VOCs). These pollutants cause smog, acid rain, and significant public health issues, such as respiratory illnesses (asthma, bronchitis), cardiovascular disease, neurological problems, and premature deaths, particularly in urban areas and communities near power plants.

* **The Solution:** Renewable energy sources produce almost no conventional air pollutants during operation, leading to cleaner air, reduced incidence of respiratory and cardiovascular diseases, and significant cost savings in healthcare.

3. **Enhancing Energy Security and Stability:**

* **The Problem:** Many nations are reliant on imported fossil fuels, often from geopolitically sensitive regions. This creates vulnerability to supply disruptions, price volatility, and can be a source of international conflict. The finite nature of fossil fuel reserves also raises concerns about future scarcity ("peak oil/gas").

* **The Solution:** Renewable resources are indigenous, widely distributed, and virtually inexhaustible. Developing robust renewable energy infrastructure reduces import dependency, insulates economies from global fuel price shocks, and provides a more stable and predictable energy supply. Decentralized renewable energy systems can also enhance grid resilience against natural disasters or cyberattacks.

4. **Fostering Economic Growth and Job Creation:**

* **The Opportunity:** The renewable energy sector is a rapidly growing industry, creating significant economic opportunities. Investments in renewable energy infrastructure lead to job creation across various sectors, including manufacturing, installation, operation, maintenance, research, and development. These jobs are often local and cannot be outsourced.

* **The Benefit:** As renewable energy technologies become more efficient and affordable, they offer lower and more stable operating costs compared to fossil fuels, which are subject to fluctuating commodity markets. This can lead to reduced energy bills for consumers and businesses.

5. **Conserving Natural Resources and Reducing Environmental Degradation:**

* **The Problem:** The extraction, processing, and transportation of fossil fuels have extensive environmental impacts. This includes habitat destruction from mining and drilling, water pollution from fracking, oil spills, and land degradation from infrastructure development. Furthermore, thermoelectric power plants (often fossil fuel-based) are significant users of freshwater for cooling.

* **The Solution:** While renewables have their own land-use considerations, their overall environmental footprint is generally lower. They eliminate the need for destructive mining and drilling operations and significantly reduce water consumption compared to traditional power generation.

6. **Driving Technological Innovation:**

* **The Momentum:** The push for renewable energy is a powerful catalyst for innovation. This includes advancements in solar panel efficiency, wind turbine design, energy storage technologies (batteries, pumped hydro), smart grid systems, and demand-side management. This continuous innovation further drives down costs and expands the applicability of renewable solutions.

In conclusion, renewable energy is fundamental to addressing the most pressing environmental challenges of our time, particularly climate change and air pollution. Beyond these ecological imperatives, it offers tangible benefits for human health, national security, and economic prosperity, paving the way for a more sustainable, resilient, and equitable future for all.

Chain-of-Thought

Renewable energy is profoundly important for a multitude of reasons, touching upon environmental, economic, social, and geopolitical aspects. Here's a breakdown of its key importance:

1. **Combating Climate Change and Environmental Degradation:**

* **Reduced Greenhouse Gas Emissions:** The primary driver of climate change is the emission of greenhouse gases (like CO₂, methane) from burning fossil fuels (coal, oil, natural gas). Renewable energy sources (solar, wind, hydro, geothermal, biomass) produce little to no greenhouse gas emissions during operation, directly mitigating global warming, extreme weather events, sea-level rise, and ocean acidification.

* **Reduced Air and Water Pollution:** Fossil fuel combustion releases harmful pollutants such as sulfur dioxide, nitrogen oxides, particulate matter, and heavy metals, leading to smog, acid rain, respiratory illnesses, and water contamination. Renewables significantly reduce these pollutants, leading to cleaner air and water.

* **Preservation of Ecosystems:** The extraction, transportation, and processing of fossil fuels often involve destructive practices like mining, drilling, and oil spills, which devastate natural habitats and biodiversity. Renewable energy generally has a much smaller ecological footprint.

2. ****Finite Resources and Energy Security:****

* ****Sustainable Resource Base:**** Fossil fuels are finite resources that will eventually run out. Renewable energy, by definition, harnesses naturally replenishing sources (sunlight, wind, water, geothermal heat, biomass), ensuring a perpetual supply of energy.

* ****Energy Independence and Security:**** Relying heavily on fossil fuels often means depending on politically volatile regions for supply, leading to energy insecurity and price volatility. Investing in domestic renewable energy sources enhances national energy independence, creates more stable and predictable energy prices, and insulates economies from global energy shocks and geopolitical tensions.

3. ****Economic Benefits and Job Creation:****

* ****Job Growth:**** The renewable energy sector is a significant and growing source of employment, creating jobs in manufacturing, installation, operation, maintenance, research, and development. These jobs are often local and cannot be easily outsourced.

* ****Decreasing Costs:**** The cost of renewable technologies, particularly solar and wind, has decreased dramatically over the past decade, making them increasingly cost-competitive with, and often cheaper than, new fossil fuel plants. This helps lower electricity bills for consumers and businesses.

* ****Innovation and Investment:**** The transition to renewable energy stimulates technological innovation, attracts investment, and fosters new industries, driving overall economic growth.

* ****Decentralization:**** Renewables enable more decentralized energy production, empowering local communities and reducing the need for expensive, centralized grid infrastructure.

4. ****Improved Public Health:****

* By reducing air and water pollution, renewable energy directly contributes to better public health outcomes. This means fewer cases of respiratory diseases, cardiovascular problems, cancers, and other ailments linked to pollution, leading to lower healthcare costs and improved quality of life.

5. ****Sustainability and Future Generations:****

* Renewable energy is fundamental to the concept of sustainable development, allowing humanity to meet its energy needs without compromising the ability of future generations to meet their own. It ensures a habitable planet and a stable resource base for millennia to come.

In essence, renewable energy is not just an alternative; it's a fundamental necessity for building a sustainable, healthy, prosperous, and secure future for all.

Evaluation:

...	Prompt Type	Accuracy	Relevance	Completeness	Clarity	Reasoning	\
0	Zero-Shot	5	5	5	5	5	
1	Few-Shot	5	5	4	5	5	
2	Role-Based	5	5	5	5	5	
3	Chain-of-Thought	5	5	5	5	5	
	Total	Output Quality (%)					
0	25	100.0					
1	24	96.0					
2	25	100.0					
3	25	100.0					
Evaluation completed successfully.							

Result:

The Gemini API successfully generated responses for each prompt type. The experiment shows that:

- **Zero-Shot Prompting** provides a direct response with minimal reasoning.
- **Few-Shot Prompting** improves consistency by learning from examples.
- **Role-Based Prompting** generates responses tailored to the assigned expertise.
- **Chain-of-Thought Prompting** produces the most detailed and logical reasoning.